

MATHEMATICAL INDUCTION

CSI 219 – Discrete Mathematics

LET'S CONSIDER
THIS PROBLEM

**Prove that you can climb
up a staircase!**



YES, YOU **HAVE** TO
MATHEMATICALLY
PROVE IT.

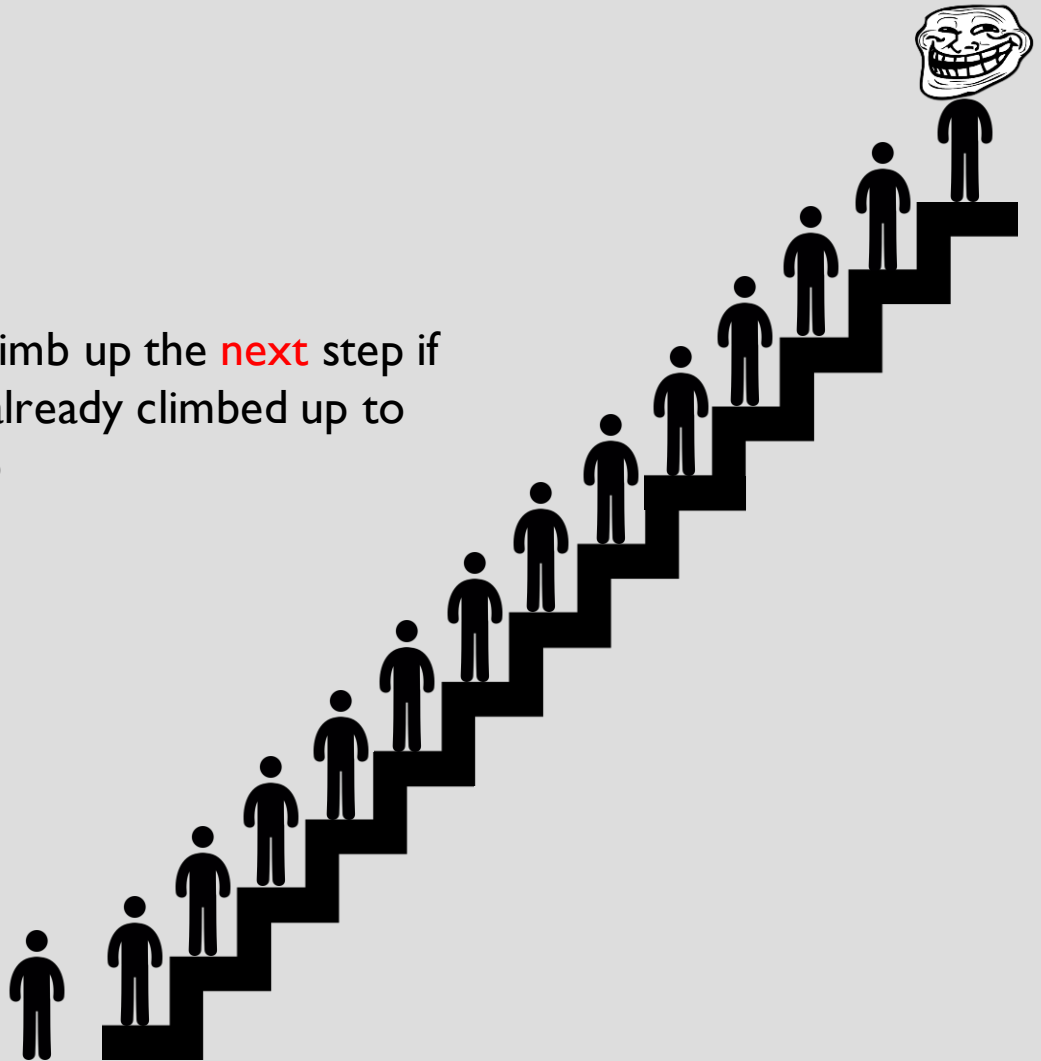


CAN YOU CLIMB UP STAIRCASES OF ANY HEIGHT?

- You only need to prove the following:
 - You can climb up the first step (this is called the base case)
 - You can climb up the **next** step if you have already climbed up to **some** arbitrary step k (this is called the inductive hypothesis)

You can climb up the **next** step if
you have already climbed up to
some step

You can climb up the first step



MATHEMATICAL INDUCTION

- A predicate is true for all values of its domain if –
 - It is true for the starting case
 - If it is true for some case, then it is true for the next case

MATHEMATICAL INDUCTION

- A predicate $P(n)$ is true for all values $n \in \mathbb{Z}^+$ if –
 - $P(1)$ is true
 - If $P(k)$ is true, then $P(k + 1)$ is true, where $k \in \mathbb{Z}^+$

PROBLEM

- Prove the following identity for finding the sum of the first n squares for all positive integer n .

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

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- Use mathematical induction to show that

$$1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$$

for all nonnegative integers n .

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